MATHEMATICS · CLASS 9 · REVISION NOTES

Formula Book

Key points and formulas for all eight chapters, with worked examples where they help most. Keep this next to your textbook and revise one chapter at a time.

CHAPTER 1

Orienting Yourself: The Use of Coordinates

Coordinate Geometry

How every point on a flat surface can be located exactly using two numbers.

KEY POINTS

- The **Cartesian plane** is formed by two perpendicular number lines: the horizontal **x-axis** and the vertical **y-axis**, crossing at the **origin O(0, 0)**.
- Any point is written as an ordered pair **(x, y)**. The first value, x, is the **abscissa**; the second, y, is the **ordinate**.
- The two axes divide the plane into four **quadrants**, numbered anti-clockwise starting from the top right.
- A point on the x-axis always has y = 0. A point on the y-axis always has x = 0.

FORMULAS

Quadrant I	(+, +)
Quadrant II	(−, +)
Quadrant III	(−, −)
Quadrant IV	(+, −)

Distance between two points $d = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$

Often introduced alongside this chapter — used constantly from here on.

CHAPTER 2

Introduction to Linear Polynomials

Polynomials

Expressions built from variables and constants, and the special case of degree 1.

KEY POINTS

- A **polynomial** in x is an expression like $a_nx^n + \ldots + a_1x + a_0$, where every power of x is a whole number.
- **Degree** = the highest power of the variable. Degree 0 = constant, degree 1 = linear, degree 2 = quadratic, degree 3 = cubic.
- A **zero** of a polynomial p(x) is a value of x for which p(x) = 0.
- A linear polynomial has exactly **one zero**. Its graph is always a straight line.

FORMULAS

Linear polynomial	$p(x) = ax + b,\ a \neq 0$
Zero of ax + b	$x = -b / a$
Remainder Theorem	$p(x) \div (x - a) \rightarrow \text{remainder} = p(a)$
Factor Theorem	$(x - a)$ is a factor of $p(x) \Leftrightarrow p(a) = 0$

SOLVED EXAMPLE

Find the zero of p(x) = 3x - 12.

Set p(x) = 0 → 3x - 12 = 0 → x = 12/3 = **4**.

REARRANGING FORMULAS (TRANSPOSITION)

- An equation stays true as long as you do the **same operation to both sides** — add, subtract, multiply, or divide both sides by the same amount.
- Moving a term across the "=" sign **flips its operation**: a term added on one side becomes subtracted on the other side (and vice versa); a term multiplying one side becomes dividing on the other side (and vice versa).

a = b + c	$\Rightarrow c = a - b$
a = b - c	$\Rightarrow c = b - a$
a = b × c	$\Rightarrow c = a / b$
a = b / c	$\Rightarrow c = b / a$
Cross-multiply: a × c = b, then divide both sides by a.	
a = b²	$\Rightarrow b = \pm\sqrt{a}$

SOLVED EXAMPLE — REARRANGING A REAL FORMULA

The area of a triangle is A = ½ × base × height. Rearrange this to find the height.

A = ½ × b × h ⇒ 2A = b × h ⇒ **h = 2A / b**

CHAPTER 3

The World of Numbers

Number Systems

The families of numbers, and the rules for handling powers and roots.

KEY POINTS

- **Natural numbers (N)**: 1, 2, 3, … **Whole numbers (W)**: 0, 1, 2, 3, … **Integers (Z)**: …, -2, -1, 0, 1, 2, …
- **Rational numbers (Q)** can be written as p/q (q ≠ 0); their decimal expansion terminates or repeats.
- **Irrational numbers** cannot be written as p/q; their decimal expansion is non-terminating and non-repeating (e.g. √2, π).
- **Real numbers (R)** = all rationals + all irrationals. Every real number has a unique point on the number line.

LAWS OF EXPONENTS

Product rule	$a^m \times a^n = a^{m+n}$
Quotient rule	$a^m \div a^n = a^{m-n}$
Power of a power	$(a^m)^n = a^{mn}$
Power of a product	$a^m \times b^m = (ab)^m$
Zero exponent	$a^0 = 1 \ \ (a \neq 0)$
Negative exponent	$a^{-n} = 1 \ / \ a^n$
Fractional exponent	$a^{1/n} = \sqrt[n]{a}$

RATIONALISATION

1 / √a $= \sqrt{a} \ / \ a$

Multiply top and bottom by √a.

1 / (√a + √b) $= (\sqrt{a} - \sqrt{b}) \ / \ (a - b)$

Multiply top and bottom by the conjugate (√a - √b).

CHAPTER 4

Exploring Algebraic Identities

Algebraic Identities — squares, cubes & factorisation

Equalities that hold true for every value of the variables — the fastest way to expand or factorise.

SQUARE IDENTITIES

Square of a sum	$(a+b)^2 = a^2 + 2ab + b^2$
Square of a difference	$(a-b)^2 = a^2 - 2ab + b^2$
Difference of squares	$a^2 - b^2 = (a+b)(a-b)$
Product of two binomials	$(x+a)(x+b) = x^2 + (a+b)x + ab$
Square of a trinomial	$(a+b+c)^2 = a^2+b^2+c^2 + 2(ab+bc+ca)$

SOLVED EXAMPLE — SQUARES

Expand $(2x + 3y)^2$ using $(a+b)^2 = a^2+2ab+b^2$, with a = 2x, b = 3y.

$= (2x)^2 + 2(2x)(3y) + (3y)^2 =$ **$4x^2 + 12xy + 9y^2$**

CUBE IDENTITIES

Cube of a sum $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
also written $a^3+b^3+3ab(a+b)$

Cube of a difference $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
also written $a^3-b^3-3ab(a-b)$

Sum of cubes $a^3+b^3 = (a+b)(a^2-ab+b^2)$

Difference of cubes $a^3-b^3 = (a-b)(a^2+ab+b^2)$

Three-term cube identity $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$
Special case: if $a+b+c = 0$, then $a^3+b^3+c^3 = 3abc$.

SOLVED EXAMPLE — CUBES

Expand $(x + 2)^3$ using $(a+b)^3 = a^3+3a^2b+3ab^2+b^3$, with $a = x$, $b = 2$.
$= x^3 + 3x^2(2) + 3x(2)^2 + 2^3 =$ **$x^3 + 6x^2 + 12x + 8$**

TRICKY QUESTIONS — USING IDENTITIES AS SHORTCUTS

MENTAL MULTIPLICATION

Find 97^2 without long multiplication.
$97^2 = (100-3)^2 = 100^2 - 2(100)(3) + 3^2 = 10000 - 600 + 9 =$ **9409**

MENTAL MULTIPLICATION — REVERSED

Find 105×95 without long multiplication.
$105 \times 95 = (100+5)(100-5) = 100^2 - 5^2 = 10000 - 25 =$ **9975**

FINDING $X^2 + 1/X^2$

If $x + 1/x = 5$, find $x^2 + 1/x^2$.
Square both sides: $(x+1/x)^2 = x^2 + 2 + 1/x^2 \rightarrow 25 = x^2+1/x^2+2 \rightarrow x^2+1/x^2 =$ **23**

FINDING $X^3 - 1/X^3$ (HARDER)

If $x - 1/x = 3$, find $x^3 - 1/x^3$.
Use $(a-b)^3 = a^3-b^3-3ab(a-b)$ with $a = x$, $b = 1/x$, so $ab = 1$:
$x^3-1/x^3 = (x-1/x)^3 + 3(x-1/x) = 3^3 + 3(3) = 27+9 =$ **36**

THE A+B+C = 0 SHORTCUT

If $a+b+c = 0$, then $a^3+b^3+c^3 = 3abc$. Check it with $a=2$, $b=3$, $c=-5$ (since $2+3+(-5)=0$):
$a^3+b^3+c^3 = 8+27+(-125) = -90$. $3abc = 3(2)(3)(-5) =$ **−90** — matches.

CUBE OF A NUMBER NEAR 1000

Find 998^3 using $(a-b)^3$ with $a = 1000$, $b = 2$.
$998^3 = 1000^3 - 3(1000)^2(2) + 3(1000)(2)^2 - 2^3$
$= 1{,}000{,}000{,}000 - 6{,}000{,}000 + 12{,}000 - 8 =$ **994,011,992**

CHAPTER 5

I'm Up and Down, and Round and Round

Circles

The theorems that govern chords, arcs and angles in a circle, plus the standard circle formulas.

KEY THEOREMS

- Equal chords of a circle subtend **equal angles** at the centre (and the converse is also true).
- The perpendicular from the centre to a chord **bisects the chord** (and the converse is also true).
- Equal chords of a circle are **equidistant** from the centre (and the converse is also true).
- The angle subtended by an arc at the centre is **double** the angle it subtends at any point on the remaining part of the circle.
- Angles in the **same segment** of a circle are equal.
- The angle in a **semicircle** is a right angle (90°).
- In a **cyclic quadrilateral**, opposite angles add up to 180° (and the converse is also true).

CIRCLE THROUGH GIVEN POINTS

- Through **one** point, infinitely many circles can be drawn.
- Through **two** points, infinitely many circles can still be drawn — their centres all lie on the **perpendicular bisector** of the segment joining the two points.
- Through **three non-collinear points** — such as the three vertices of a triangle — exactly **one** circle can be drawn. This is the triangle's **circumcircle**.
- The centre of this circle (the **circumcentre**) is where the perpendicular bisectors of all three sides meet; it is equidistant from all three vertices.
- If the triangle is **right-angled**, the circumcentre is simply the **midpoint of the hypotenuse**, and the hypotenuse is a diameter — a direct consequence of "angle in a semicircle = 90°" above.
- If the three points are **collinear** (lie on a straight line), no circle can pass through all three.

CIRCUMRADIUS & INRADIUS OF A TRIANGLE

Circumradius (any triangle) $R = (a \times b \times c) / (4 \times \text{Area})$

a, b, c = the triangle's three sides; Area from Heron's formula (Chapter 6).

Circumradius (right triangle) $R = \text{hypotenuse} / 2$

Inradius (circle touching all 3 sides) $r = \text{Area} / s$

s = semi-perimeter = (a+b+c)/2.

SOLVED EXAMPLE — CIRCLE THROUGH A RIGHT TRIANGLE'S VERTICES

A right triangle has legs 3 cm and 4 cm, so its hypotenuse = 5 cm. Find the radius of the circle passing through all three vertices.

The hypotenuse of a right triangle is always a diameter of its circumcircle, so R = hypotenuse / 2 = 5/2 = **2.5 cm**.

SOLVED EXAMPLE — CIRCLE THROUGH ANY TRIANGLE'S VERTICES

A triangle has sides 5 cm, 6 cm and 7 cm. Find the radius of the circle passing through its three vertices.

From Heron's formula: Area = $\sqrt{216} \approx 14.7$ cm².

R = (5 × 6 × 7) / (4 × 14.7) = 210 / 58.8 ≈ **3.57 cm**

CIRCLE FORMULAS

Circumference	$C = 2\pi r = \pi d$
Area	$A = \pi r^2$
Area of a semicircle	$A = \frac{1}{2}\pi r^2$
Length of an arc	$l = (\theta/360) \times 2\pi r$
θ = angle subtended at the centre, in degrees.	
Area of a sector	$A = (\theta/360) \times \pi r^2$
Diameter	$d = 2r$
the longest possible chord of a circle.	

CHAPTER 6

Measuring Space: Perimeter and Area

Mensuration — Perimeter, Area & Heron's Formula

How to measure the boundary and the surface of flat shapes.

PERIMETER & AREA — FLAT SHAPES

Rectangle	$P = 2(l+b), \quad A = l \times b$
Square	$P = 4a, \quad A = a^2$
Triangle	$A = \frac{1}{2} \times \text{base} \times \text{height}$
Equilateral triangle	$A = (\sqrt{3} / 4)\, a^2$
Parallelogram	$A = \text{base} \times \text{height}$
Rhombus	$A = \frac{1}{2} \times d_1 \times d_2$
d_1, d_2 = the two diagonals.	
Trapezium	$A = \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$

HERON'S FORMULA — AREA FROM THREE SIDES

Semi-perimeter	$s = (a+b+c) / 2$
Area of any triangle	$A = \sqrt{[s(s-a)(s-b)(s-c)]}$
Useful when you know all three sides but not the height.	

SOLVED EXAMPLE — HERON'S FORMULA

A triangle has sides 5 cm, 6 cm and 7 cm. Find its area.

$s = (5+6+7)/2 = 9$. $A = \sqrt{[9(9-5)(9-6)(9-7)]} = \sqrt{(9\times4\times3\times2)} = \sqrt{216} \approx$ **14.7 cm²**

BONUS: SURFACE AREA & VOLUME — SOLIDS

Cube	$SA = 6a^2$, $V = a^3$
Cuboid	$SA = 2(lb+bh+hl)$, $V = l \times b \times h$
Cylinder	$CSA = 2\pi rh$, $V = \pi r^2 h$
Cone l (slant height) = $\sqrt{(r^2+h^2)}$	$CSA = \pi rl$, $V = \frac{1}{3}\pi r^2 h$
Sphere	$SA = 4\pi r^2$, $V = \frac{2}{3}\pi r^3$
Hemisphere	$CSA = 2\pi r^2$, $V = (2/3)\pi r^3$

CHAPTER 7

The Mathematics of Maybe: Introduction to Probability

Probability

How likely an event is, based on what actually happens when you try it.

KEY POINTS

- Class 9 probability is **experimental (empirical)**: based on the results of trials you actually carry out or observe.
- Probability always lies between 0 and 1, inclusive.
- P(E) = 0 means E is an **impossible event**; P(E) = 1 means E is a **sure (certain) event**.

FORMULAS

Probability of an event	$P(E) = (\text{trials where E happened}) / (\text{total trials})$
Range	$0 \le P(E) \le 1$
Complement rule	$P(\text{not E}) = 1 - P(E)$

SOLVED EXAMPLE

A coin is tossed 30 times; heads come up 17 times. Find the experimental probability of heads.

P(heads) = 17/30 ≈ **0.57**

CHAPTER 8

Predicting What Comes Next: Exploring Sequences and Progressions

Sequences & Progressions

Patterns of numbers that follow a fixed rule from one term to the next.

KEY POINTS

- A **sequence** is an ordered list of numbers (called terms) that follows a pattern.
- An **Arithmetic Progression (AP)** has a constant **common difference** (d) between consecutive terms: a, a+d, a+2d, ...

• A **Geometric Progression (GP)** has a constant **common ratio** (r) between consecutive terms: a, ar, ar², …

ARITHMETIC PROGRESSION (AP)

nth term	$a_n = a + (n-1)d$
Sum of n terms	$S_n = n/2\,[2a + (n-1)d]$

also written $S_n = n/2\,(a + l)$, where l = last term.

GEOMETRIC PROGRESSION (GP)

nth term	$a_n = a \cdot r^{n-1}$
Sum of n terms (r ≠ 1)	$S_n = a(r^n-1) \,/\, (r-1)$
Sum to infinity	$S_\infty = a \,/\, (1-r)$

only valid when $-1 < r < 1$.

SOLVED EXAMPLE — AP

Find the 10th term and the sum of the first 10 terms of 3, 7, 11, 15, …

a = 3, d = 4. $a_{10} = 3 + (10-1)(4) =$ **39**. $S_{10} = 10/2\,[2(3) + 9(4)] = 5(6+36) =$ **210**

QUICK REFERENCE

All Formulas at a Glance

One last scan before the exam — the most-used formula from every chapter.

COORDINATES & NUMBERS

Distance $d = \sqrt{[(x_2-x_1)^2+(y_2-y_1)^2]}$ · $a^m \times a^n = a^{m+n}$ · $a^m \div a^n = a^{m-n}$ · $(a^m)^n = a^{mn}$ · $a^{-n} = 1/a^n$

IDENTITIES

$(a+b)^2 = a^2+2ab+b^2$ · $(a-b)^2 = a^2-2ab+b^2$ · $a^2-b^2 = (a+b)(a-b)$ · $(a+b)^3 = a^3+3a^2b+3ab^2+b^3$ · $(a-b)^3 = a^3-3a^2b+3ab^2-b^3$ · $a^3+b^3 = (a+b)(a^2-ab+b^2)$ · $a^3-b^3 = (a-b)(a^2+ab+b^2)$

CIRCLES & MENSURATION

Circumference $= 2\pi r$ · **Area of circle** $= \pi r^2$ · **Sector area** $= (\theta/360)\pi r^2$ · **Heron's** $A = \sqrt{[s(s-a)(s-b)(s-c)]}$ · **Equilateral** $\triangle = (\sqrt{3}/4)a^2$ · **Cylinder V** $= \pi r^2 h$ · **Cone V** $= \frac{1}{3}\pi r^2 h$ · **Sphere V** $= \frac{2}{3}\pi r^3$

PROBABILITY & PROGRESSIONS

P(E) = favourable trials / total trials · **P(not E)** = 1-P(E) · **AP nth term** $= a+(n-1)d$ · **AP sum** $= n/2[2a+(n-1)d]$ · **GP nth term** $= ar^{n-1}$ · **GP sum** $= a(r^n-1)/(r-1)$

Revise one chapter a day rather than all eight at once — and redo the solved examples yourself before checking them.

You've got this.